

Types of Cyber Threats

A detailed guide for the CyberRAG cybersecurity knowledge base

1. Understanding Cyber Threats

A cyber threat is any potential event, actor, action, or circumstance capable of harming a digital system, network, application, organization, or user. The harm may involve unauthorized access, theft, destruction, modification, disruption, surveillance, fraud, or loss of privacy. Cyber threats can originate from external attackers, insiders, criminal groups, automated systems, compromised devices, or accidental human actions.

2. Important Security Concepts

Threat: a potential cause of harm.

Vulnerability: a weakness in software, hardware, configuration, process, or human behavior.

Exploit: a technique that takes advantage of a vulnerability.

Attack: an attempt to compromise or disrupt a target.

Risk: the potential impact and likelihood of a threat successfully exploiting a vulnerability.

Security control: a safeguard used to prevent, detect, respond to, or recover from threats.

3. Malware Threats

Malware, short for malicious software, is software designed to perform unwanted or harmful actions. Depending on its type, malware may steal information, spy on users, encrypt data, destroy files, provide unauthorized access, or spread through networks. Common categories include viruses, worms, Trojans, ransomware, spyware, rootkits, keyloggers, bots, backdoors, adware, scareware, fileless malware, cryptojacking malware, and logic bombs.

How it commonly enters: malicious attachments, unsafe downloads, compromised websites, vulnerable services, removable media, or deceptive software.

Impact: data loss, financial damage, downtime, credential theft, privacy loss, and unauthorized access.

Defenses: patching, endpoint protection, application control, backups, least privilege, monitoring, and safe user behavior.

4. Phishing and Social Engineering

Social engineering attacks exploit human trust, urgency, fear, curiosity, or authority. Phishing commonly uses fraudulent messages or websites to convince a victim to reveal credentials, open an attachment, send money, or perform another unsafe action.

Variants: general email phishing, spear phishing aimed at a specific target, whaling aimed at senior individuals, smishing through SMS, vishing through voice calls, and business email compromise involving impersonation of trusted business contacts.

Warning signs: unexpected urgency, suspicious domains, unusual payment requests, mismatched links, unfamiliar attachments, or requests to bypass normal procedures.

Defenses: verification through a separate trusted channel, awareness training, email filtering, MFA, and careful link inspection.

5. Credential and Authentication Threats

These threats target usernames, passwords, authentication cookies, tokens, recovery mechanisms, or other identity information. Attackers may attempt to guess credentials, reuse leaked credentials, steal them through

phishing, or abuse weak authentication processes.

Examples: brute-force attacks, dictionary attacks, password spraying, credential stuffing, session theft, and phishing-based credential theft.

Impact: account takeover and access to data or internal systems.

Defenses: unique passwords, password managers, MFA, rate limiting, login monitoring, secure recovery processes, and alerts for unusual authentication activity.

6. Denial-of-Service and Distributed Denial-of-Service

A denial-of-service attack attempts to reduce or completely prevent access to a service. It can consume bandwidth, processing power, memory, application resources, or connection capacity. A distributed denial-of-service attack uses many sources, often compromised devices, making filtering more difficult.

Impact: website outages, business interruption, financial loss, and disruption of critical services.

Defenses: rate limiting, traffic filtering, redundancy, load balancing, content delivery infrastructure, capacity planning, and DDoS mitigation services.

7. Web Application Threats

Web applications can contain weaknesses in input handling, authentication, authorization, session management, configuration, or third-party components. Attackers may attempt to manipulate the application to access data or perform unauthorized actions.

Examples: injection flaws, cross-site scripting, broken access control, authentication failures, insecure design, security misconfiguration, vulnerable dependencies, and server-side request manipulation.

Defenses: secure coding, parameterized queries, input validation, output encoding, authorization checks, dependency updates, security testing, logging, and code review.

8. Man-in-the-Middle and Communication Interception

A communication interception threat occurs when an attacker gains the ability to observe or potentially alter data exchanged between parties. This can be especially dangerous when communication lacks proper encryption or certificate validation.

Possible impact: credential theft, data exposure, session compromise, or manipulation of transmitted information.

Defenses: HTTPS and modern TLS, correct certificate validation, secure network configuration, VPNs where appropriate, and avoiding untrusted networks for sensitive activity.

9. Network Spoofing and Redirection Threats

Spoofing attempts to impersonate or falsify an identity, address, or trusted resource. Network-related threats can include IP spoofing, DNS manipulation, ARP spoofing, rogue access points, and deceptive redirection.

Impact: traffic interception, misdirection, credential theft, or unauthorized access.

Defenses: secure DNS practices, network segmentation, switch security features, monitoring, strong Wi-Fi security, and detection of unusual network behavior.

10. Insider Threats

An insider threat involves a person or account with legitimate access that intentionally misuses access or accidentally creates a security problem. Insiders may be employees, administrators, contractors, or partners.

Types: malicious insiders acting intentionally; negligent insiders making unsafe mistakes; and compromised insiders whose legitimate accounts have been taken over.

Defenses: least privilege, access reviews, separation of duties, monitoring, logging, clear offboarding, and security awareness.

11. Advanced Persistent Threats

An advanced persistent threat generally describes a targeted intrusion in which an attacker seeks to establish and maintain access over an extended period. The objective may involve intelligence gathering, data theft, long-term surveillance, or strategic disruption.

Typical characteristics: persistence, stealth, multiple stages, and adaptation to defensive controls. A long-term intrusion may involve initial access, persistence, privilege escalation, credential access, discovery, lateral movement, collection, and data exfiltration.

Defenses: layered monitoring, identity security, segmentation, threat hunting, logging, incident response, and rapid containment.

12. Zero-Day Threats

A zero-day threat involves a previously unknown or not-yet-remediated vulnerability. A zero-day attack attempts to exploit that weakness before a reliable fix is available or deployed.

Risk: organizations may not have a specific patch or signature available.

Defenses: defense in depth, behavior-based detection, application isolation, least privilege, segmentation, rapid patch deployment when available, and reduction of unnecessary exposed services.

13. Supply-Chain Threats

A supply-chain attack compromises a trusted component, vendor, dependency, update process, service provider, or distribution mechanism. The attacker may target the supply chain because one compromise can potentially affect many downstream users.

Examples: compromised software dependencies, malicious or altered updates, stolen signing credentials, compromised vendor accounts, and insecure third-party services.

Defenses: supplier assessment, software inventories, dependency management, integrity verification, controlled updates, monitoring, and incident response planning.

14. Cloud Security Threats

Cloud environments can be affected by exposed storage, weak identity controls, excessive permissions, leaked access keys, insecure APIs, account compromise, and inadequate monitoring. Many cloud incidents involve configuration or identity-management errors rather than a failure of the cloud provider itself.

Defenses: least-privilege IAM, MFA, secure configuration reviews, secrets management, logging, encryption, network controls, continuous monitoring, and clear understanding of shared security responsibilities.

15. IoT and Embedded Device Threats

Internet-connected cameras, sensors, smart appliances, industrial devices, and other embedded systems may have weak passwords, outdated firmware, insecure services, or limited security monitoring. Large numbers of vulnerable devices can also be abused as part of botnets.

Defenses: change default credentials, update firmware, disable unnecessary services, isolate devices on separate networks, and monitor unusual connections.

16. Mobile Security Threats

Mobile threats include malicious or counterfeit applications, excessive permissions, spyware, SMS phishing, insecure Wi-Fi use, device theft, and exploitation of outdated operating systems or applications.

Defenses: install applications from trusted sources, review permissions, keep the operating system updated, use screen locks and device encryption, enable account protection, and avoid interacting with suspicious messages.

17. Data Breaches and Data Theft

A data breach occurs when information is accessed, exposed, copied, altered, or stolen without authorization. Targets may include personal data, passwords, financial information, business records, intellectual property, and authentication secrets.

Common causes: stolen credentials, vulnerable applications, malware, insider misuse, misconfigured storage, and insecure third-party access.

Defenses: access control, encryption, data classification, monitoring, secure backups, DLP where appropriate, and incident response.

18. Ransom and Extortion Threats

Digital extortion can involve encryption of systems, theft of sensitive data, threats to publish information, or threats to disrupt services. Modern incidents may combine multiple forms of pressure rather than relying only on file encryption.

Defenses: resilient backups, identity protection, segmentation, monitoring, tested incident response plans, and recovery exercises. Organizations should follow applicable legal, regulatory, and incident-handling procedures.

19. Cryptographic and Certificate Threats

Cryptographic security can fail because of weak algorithms, poor implementation, insecure key storage, inadequate random number generation, improper certificate validation, or weak passwords protecting sensitive keys.

Defenses: use current recommended cryptographic standards, protect and rotate keys appropriately, validate certificates correctly, avoid obsolete protocols, and use established cryptographic libraries rather than creating custom implementations.

20. Physical and Hardware Threats

Cybersecurity also depends on physical protection. Lost or stolen devices, unauthorized access to servers, device tampering, rogue USB devices, and unauthorized entry into restricted areas can lead to digital compromise.

Defenses: device encryption, secure storage, access badges, asset management, screen locks, secure disposal, and controls over removable media and physical access.

21. Misconfiguration and Human Error

Not every cyber threat begins with sophisticated malware. Incorrect permissions, publicly exposed services, default credentials, unpatched software, accidental data sharing, and insecure cloud settings can create serious security risks.

Defenses: secure configuration baselines, automated checks, change management, peer review, patch management, least privilege, and regular security assessments.

22. Recognizing a Possible Cyber Incident

Warning signs may include unusual login locations, repeated authentication failures, unexpected password-reset requests, unexplained administrator accounts, suspicious processes, unusual network traffic, unauthorized configuration changes, unexpected encryption of files, abnormal cloud activity, or alerts from security tools. A single indicator does not always prove compromise, but unusual events should be investigated according to the organization's procedures.

23. General Cyber Threat Prevention Strategy

1. Maintain an accurate inventory of important systems, applications, accounts, and data.
2. Keep operating systems, applications, firmware, and security tools updated.
3. Use strong unique passwords and multi-factor authentication.
4. Apply least privilege and regularly review access.
5. Maintain tested backups and protect them from unauthorized modification.
6. Use network segmentation and secure configurations.
7. Monitor important authentication, endpoint, network, cloud, and application activity.
8. Train users to recognize phishing and verify unusual requests.
9. Test security controls and remediate discovered vulnerabilities.
10. Maintain and regularly exercise incident response and recovery procedures.

24. Summary

Cyber threats are diverse and often interconnected. A phishing message can lead to credential theft, stolen credentials can lead to unauthorized access, and unauthorized access can result in malware deployment, data theft, ransomware, or long-term persistence. Effective cybersecurity therefore requires defense in depth: preventive controls, detection and monitoring, incident response, recovery, and continuous improvement.

CyberRAG Knowledge Base Note: Each threat category in this document is separated into a focused section with definitions, examples, impacts, warning signs, and defensive concepts. This structure is useful for retrieval because questions about a specific threat can be matched to a dedicated knowledge section.